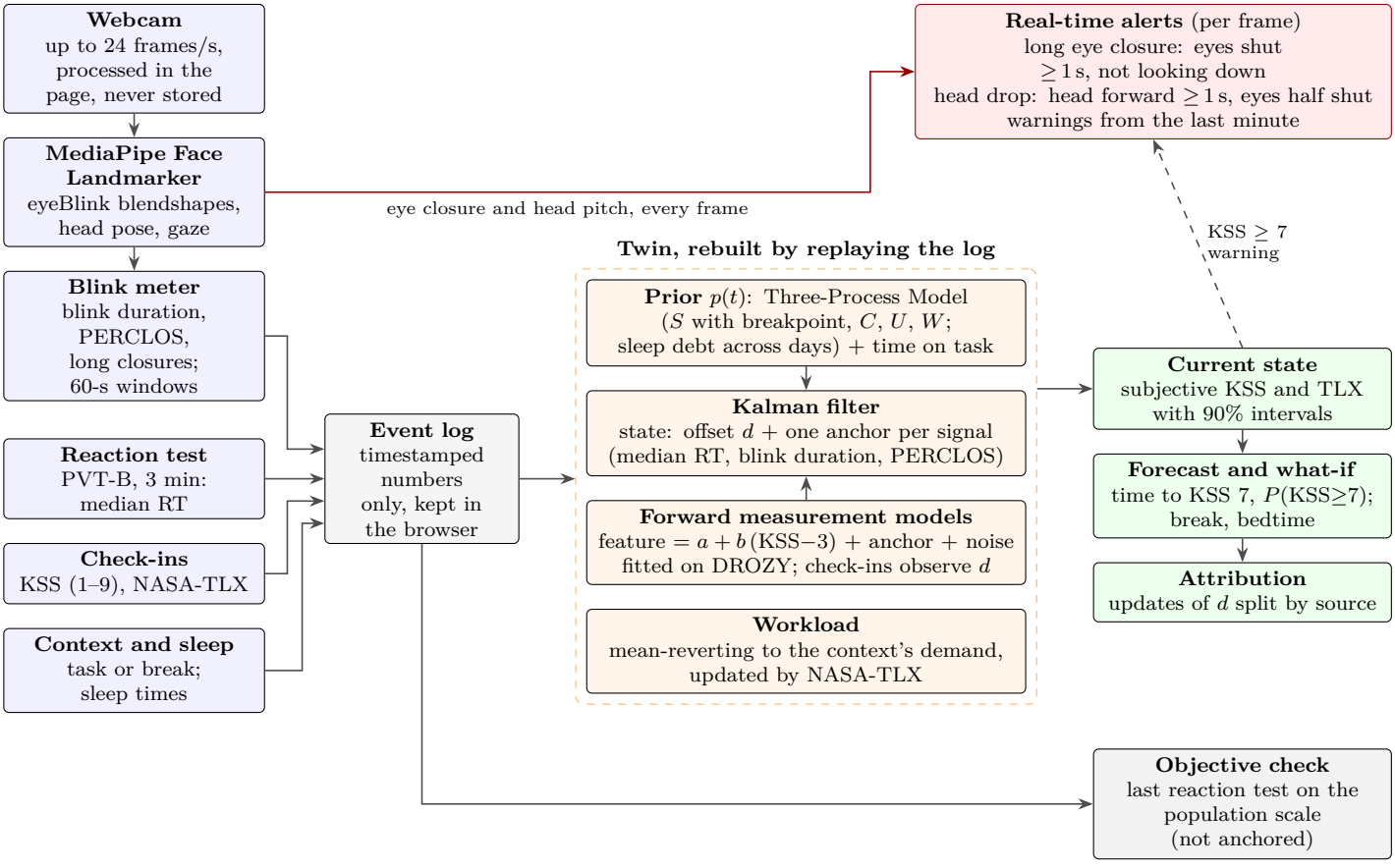

Webcam
up to 24 frames/s, processed in the page, never stored
MediaPipe Face Landmarker
eyeBlink blendshapes, head pose, gaze
Blink meter
blink duration, PERCLOS, long closures; 60-s windows
Reaction test
PVT-B, 3 min: median RT
Check-ins
KSS (1–9), NASA-TLX
Context and sleep
task or break; sleep times
Event log
timestamped numbers only, kept in the browser
eye closure and head pitch, every frame
Real-time alerts (per frame)
long eye closure: eyes shut ≥ 1 s, not looking down
head drop: head forward ≥ 1 s, eyes half shut
warnings from the last minute
KSS ≥ 7 warning
Twin, rebuilt by replaying the log
Prior p(t): Three-Process Model (S with breakpoint, C, U, W; sleep debt across days) + time on task
Kalman filter
state: offset d + one anchor per signal (median RT, blink duration, PERCLOS)
Forward measurement models
feature = a + b (KSS−3) + anchor + noise fitted on DROZY; check-ins observe d
Workload
mean-reverting to the context's demand, updated by NASA-TLX
Current state
subjective KSS and TLX with 90% intervals
Forecast and what-if
time to KSS 7, P(KSS≥7); break, bedtime
Attribution
updates of d split by source
Objective check
last reaction test on the population scale (not anchored)